

| Question | Answer           | Mark | Mark scheme                                                                                                                                                                                                                                                                                                       | Additional guidance           |
|----------|------------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| 22       | $x^2 + y^2 = 80$ | P1   | for process to find gradient of tangent eg $\frac{10-0}{0--20}$ ( $=\frac{1}{2}$ )<br><b>or</b> for $20^2 + 10^2 (= 500)$<br><b>or</b> start to method to find angle between tangent and $x$ axis,<br>eg $\tan\theta = \frac{10}{20}$                                                                             |                               |
|          |                  | P1   | for process to find gradient of normal/radius eg $\frac{-1}{\text{"0.5"}}$ ( $=-2$ )<br><b>or</b> for $\sqrt{20^2 + 10^2}$ or $\sqrt{500}$ or 22.36... or 22.4<br><b>or</b> completes process to find angle between tangent and $x$ axis.<br>eg $\theta = \tan^{-1}\left(\frac{10}{20}\right)$ ( $=26.565\dots$ ) |                               |
|          |                  | P1   | for equation of tangent eg $y = \text{"0.5"}x + 10$ oe<br>or for equation of radius eg $y = \text{"-2"}x$ oe<br><b>or</b> for using similar triangles eg $\frac{r}{10} = \frac{20}{\text{"}\sqrt{500}\text{"}}$<br><b>or</b> for $\sin(\text{"26.565..."}) = \frac{r}{20}$                                        |                               |
|          |                  | P1   | for process to find the $x$ coordinate eg $\text{"0.5"}x + 10 = \text{"-2"}x$ ( $x = -4$ )<br><b>or</b> for $r = \frac{20}{\text{"}\sqrt{500}\text{"}} \times 10$<br><b>or</b> $r = 20 \times \sin(\text{"26.565..."})$                                                                                           |                               |
|          |                  | A1   | oe                                                                                                                                                                                                                                                                                                                | Accept $(4\sqrt{5})^2$ for 80 |

**Questions that specifically require working:** In general, candidates who do not show working on this type of question will get no marks – full details will be given in the mark scheme for each individual question.